

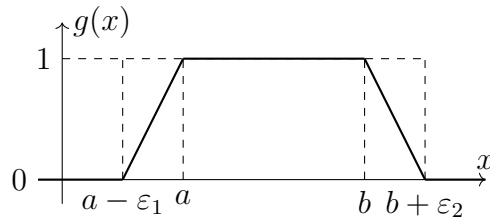
Lem aprox indicatriz continua

Lema 1. Sean $a, b \in \mathbb{R}$ con $a < b$ y sean $\varepsilon_1, \varepsilon_2 > 0$

$$\implies \exists g \in \mathcal{C}_c(\mathbb{R}) : \forall x \in \mathbb{R} : \mathbb{1}_{[a,b]}(x) \leq g(x) \leq \mathbb{1}_{[a-\varepsilon_1, b+\varepsilon_2]}(x).$$

Demostración: Basta con tomar g dada por $\forall x \in [a - \varepsilon_1, b + \varepsilon_2]$:

$$g(x) = \begin{cases} \frac{x - (a - \varepsilon_1)}{\varepsilon_1} & a - \varepsilon_1 < x < a, \\ 1 & a \leq x \leq b, \\ \frac{(b + \varepsilon_2) - x}{\varepsilon_2} & b < x < b + \varepsilon_2, \end{cases}$$



y $g(x) = 0$ fuera de $[a - \varepsilon_1, b + \varepsilon_2]$. ■

Referenciado en

- lem-aprox-indicatriz-compacto-abierto-continua

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